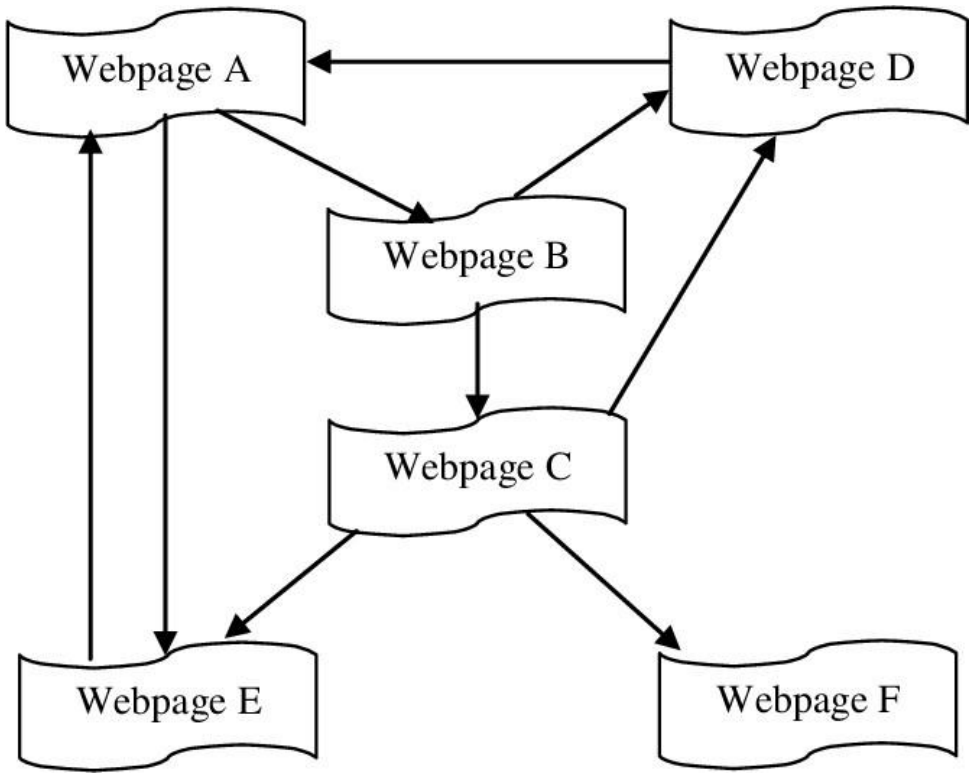
Page Rank Algorithm

Dr. Abhilasha Chaudhuri

PageRank: Intuition



Navadiya and Garg [2011]

- Outgoing hyperlink from a page is termed as out-edge or forward link
- Incoming hyperlink to a page from the second one is termed as an in-edge or backward link
- With every forward link a page establishes,
  - it transfers some of its importance/rank influence to the forward page
- If a highly important node points to a lesser important one,
  - there is an enhancement in the status of the latter node
- Importance of each node is determined by its in-edges/backward links

PageRank: Simple Ranking

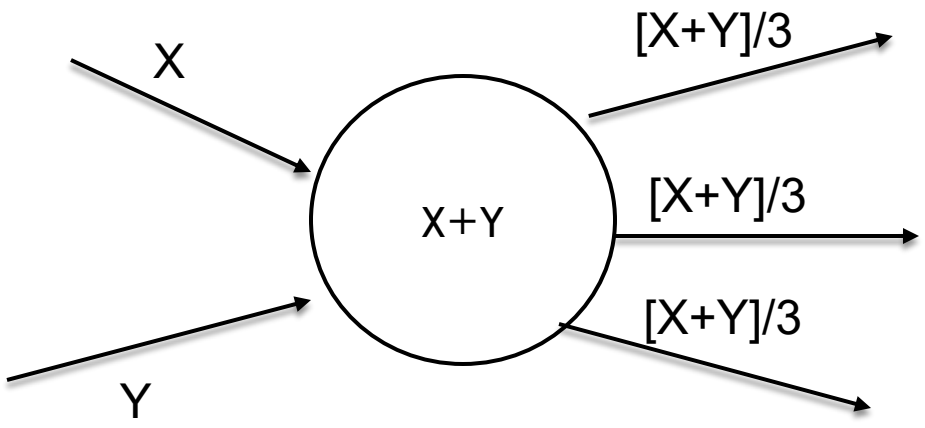
For a node w, let F_w be the set of nodes that w points to (Forward links) and B_w be the set of node that points to w (Backward links). Further, let $N_w = |F_w|$, the number of forward links from w. Then, the simple ranking of w, denoted $R(w)$, is given by

$$R(w) = \sum_{b \in B_w} \frac{R(b)}{N_b}$$

- The underlying web graph is assumed to be a connected component
- There could be pages that neither refer to any other page nor are referred to by any other page
- In a scenario where no hyperlinks exist in the network
  - Each page is assumed to be equally (un)important with a uniform rank given by

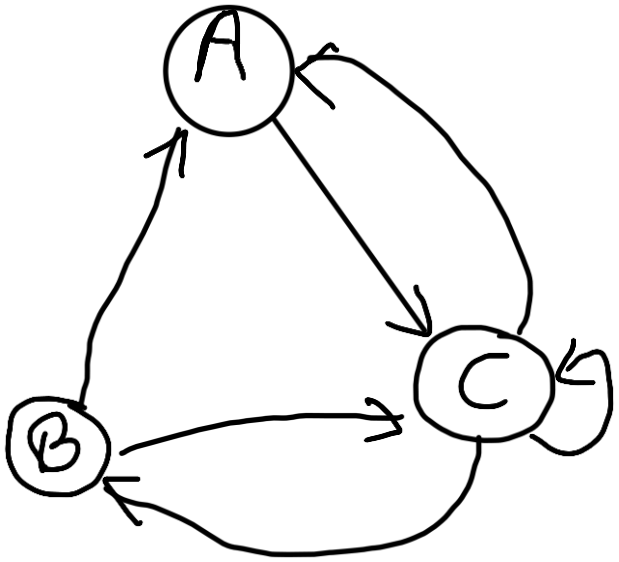
$$R(p) = \frac{1}{\#Webpages}$$

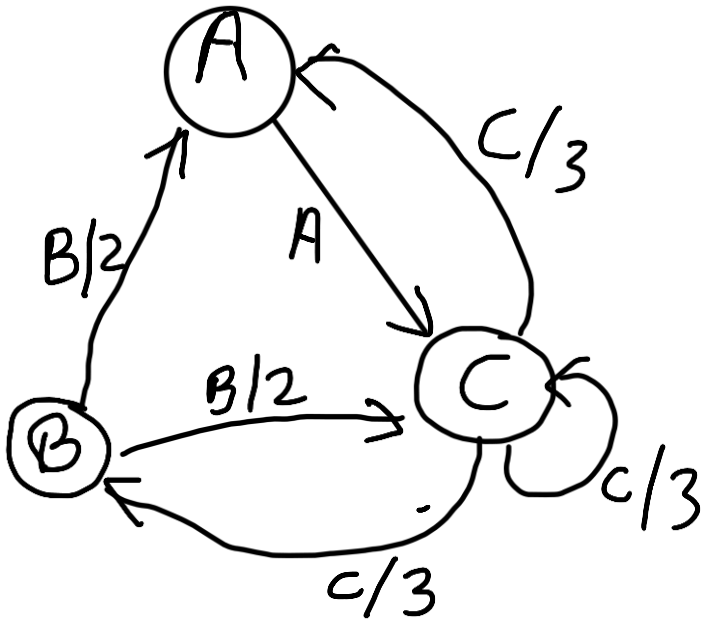
PageRank: Simple Ranking



- Importance of a web page is the sum of all the importance of its incoming links.
- The web page distributes its importance equally to all its outgoing links.

PageRank: Simple Ranking





$$A = \frac{B}{2} + \frac{C}{3}$$

$$B = \frac{C}{3}$$

$$C = A + \frac{B}{2} + \frac{C}{3}$$

$$A + B + C = 1$$

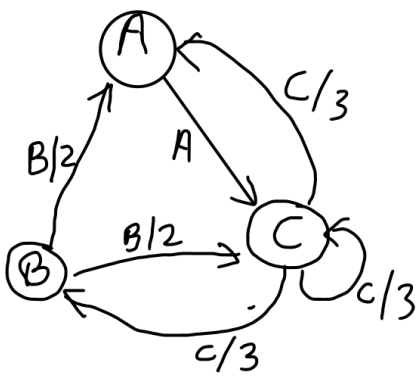
PageRank: Simple Ranking

Adjacency Matrix

Source nodes

nodes to used

	A	B	C
A	0	1	1
B	0	0	1
C	1	1	1
sum	1	2	3



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$$A + B + C =$$

PageRank: Simple Ranking

Adjacency Matrix

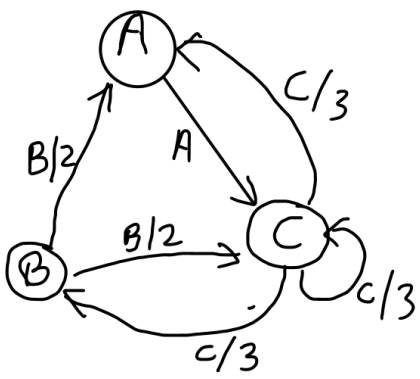
Source nodes

Destination nodes

	A	B	C
A	0	1	1
B	0	0	1
C	1	1	1
Sum	1	2	3

Transition Matrix

	A	B	C
A	0	1/2	1/3
B	0	0	1/3
C	1	1/2	1/3



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$$A + B + C =$$

PageRank: Simple Ranking

Adjacency Matrix

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Destination nodes

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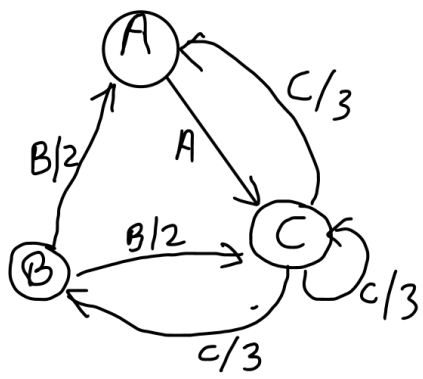


$$A = \frac{B}{2} + \frac{C}{3}$$

$$B = \frac{C}{3}$$

$$C = A + \frac{B}{2} + \frac{C}{3}$$

$$A + B + C =$$

Transition Matrix

	A	B	C
A	0	1/2	1/3
B	0	0	1/3
C	1	1/2	1/3

Rank Vector

1/3	A
1/3	B
1/3	C

- Rows of the transition matrix and Adjacency matrix represent the incoming links.
- The columns represents the outgoing links.
- A vector that contains importance of all web pages in the network is called a "Rank vector"
- Rank vector is a column vector where each cell represents the importance of the corresponding web page

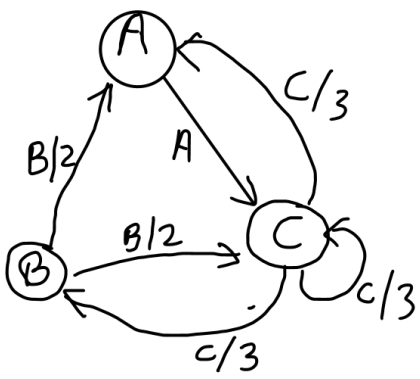
PageRank: Simple Ranking

Adjacency Matrix

Source nodes

Destination nodes

	A	B	C
A	0	1	1
B	0	0	1
C	1	1	1
Sum	1	2	3



$$A = \frac{B}{2} + \frac{C}{3}$$

$$B = \frac{C}{3}$$

$$C = A + \frac{B}{2} + \frac{C}{3}$$

$$A + B + C =$$

Transition Matrix

	A	B	C
A	0	1/2	1/3
B	0	0	1/3
C	1	1/2	1/3

Rank Vector

1/3	A
1/3	B
1/3	C

Consider nodes A, B and C :
Importance of node A = $0 \times \frac{1}{3} + \frac{1}{2} \times \frac{1}{3} + \frac{1}{3} \times \frac{1}{3}$

$$B = 0 \times \frac{1}{3} + 0 \times \frac{1}{3} + \frac{1}{3} \times \frac{1}{3}$$

$$C = 1 \times \frac{1}{3} + \frac{1}{2} \times \frac{1}{3} + \frac{1}{3} \times \frac{1}{3}$$

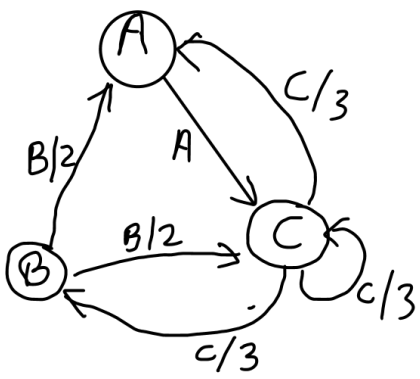
PageRank: Simple Ranking

Adjacency Matrix

Source nodes

Destination nodes

	A	B	C
A	0	1	1
B	0	0	1
C	1	1	1
Sum	1	2	3



$$A = \frac{B}{2} + \frac{C}{3}$$

$$B = \frac{C}{3}$$

$$C = A + \frac{B}{2} + \frac{C}{3}$$

$$A + B + C =$$

Transition Matrix

	A	B	C
A	0	1/2	1/3
B	0	0	1/3
C	1	1/2	1/3

Rank Vector

1/3	A
1/3	B
1/3	C

Importance of node A = $0 \times \frac{1}{3} + \frac{1}{2} \times \frac{1}{3} + \frac{1}{3} \times \frac{1}{3} = 5/18$

$$B = 0 \times \frac{1}{3} + 0 \times \frac{1}{3} + \frac{1}{3} \times \frac{1}{3} = 1/9$$

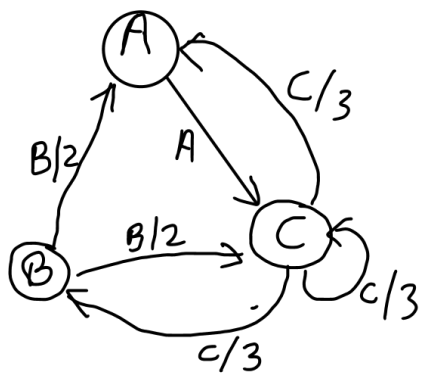
$$C = 1 \times \frac{1}{3} + \frac{1}{2} \times \frac{1}{3} + \frac{1}{3} \times \frac{1}{3} = 11/18$$

PageRank: Simple Ranking

Adjacency Matrix

Source nodes

Destination nodes	A	B	C
A	0	1	1
B	0	0	1
C	1	1	1
Sum	1	2	3



$$A = \frac{B}{2} + \frac{C}{3}$$

$$B = \frac{C}{3}$$

$$C = A + \frac{B}{2} + \frac{C}{3}$$

$$A + B + C =$$

Transition Matrix

	A	B	C
A	0	1/2	1/3
B	0	0	1/3
C	1	1/2	1/3

New Rank Vector

5/18	A
1/9	B
11/18	C

$$= 0 \times \frac{5}{18} + \frac{1}{2} \times \frac{1}{9} + \frac{1}{3} \times \frac{11}{18} = 0.259$$

$$= 0 \times \frac{5}{18} + 0 \times \frac{1}{9} + \frac{1}{3} \times \frac{11}{18} = 0.2037$$

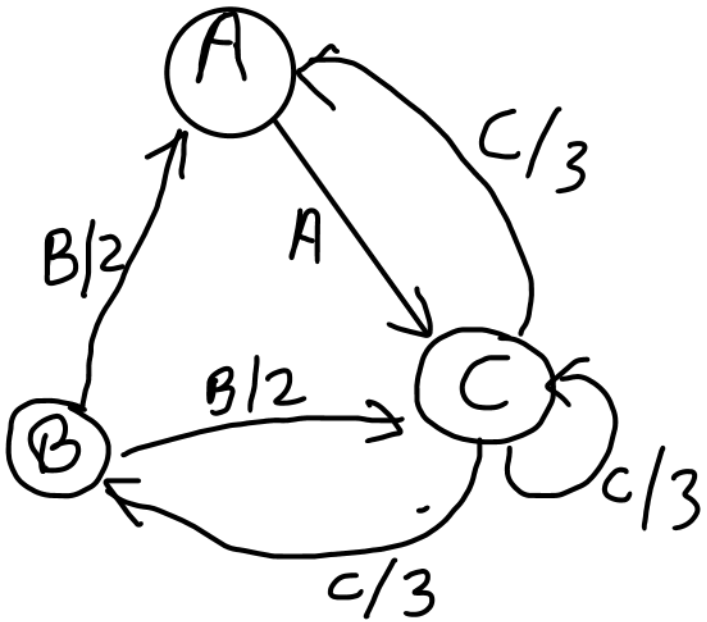
$$= 1 \times \frac{5}{18} + \frac{1}{2} \times \frac{1}{9} + \frac{1}{3} \times \frac{11}{18} = 0.537$$

PageRank: Simple Ranking

Iterations →

1/3	5/18	0.259	0.28086	0.269547	-------	0.272727
1/3	1/9	0.2037	0.174012	0.18004115		0.18181818
1/3	11/18	0.537	0.54012	0.55041152		0.54545454

PageRank: Simple Ranking



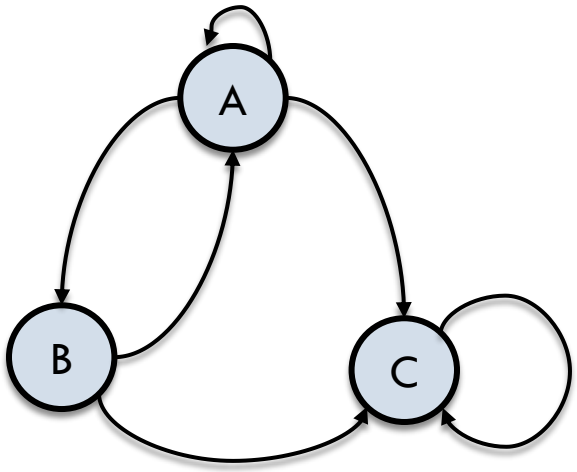
Network

0.272727
0.18181818
0.54545454

Page rank vector

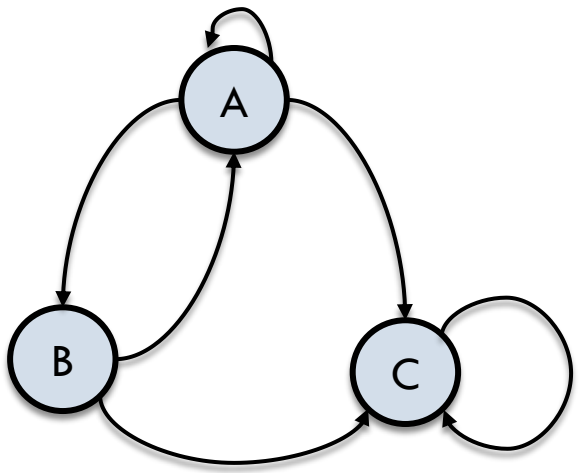
Spider Trap

Spider Trap



- The node C do not have any outgoing links except a self-loop.
- It means that C will not share its importance with any other node.
- Therefore, eventually after many iterations the page rank of C will become 1 and other nodes will have page rank value of zero.
- This problem is called spider trap.

Spider Trap



Adjacency Matrix

	A	B	C
A	1	1	0
B	1	0	0
C	1	1	1

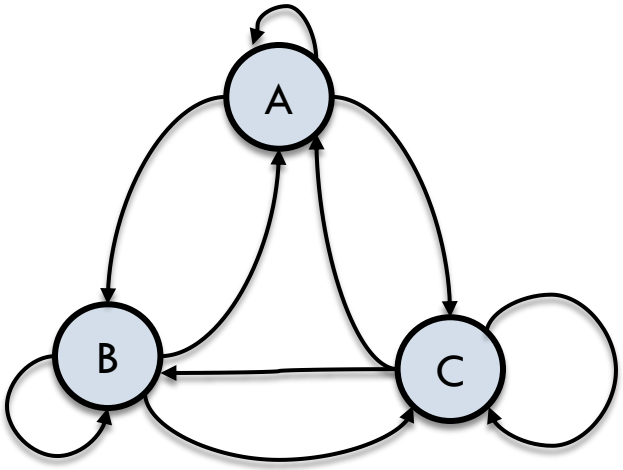
Transition Matrix

	A	B	C
A	1/3	1/2	0
B	1/3	0	0
C	1/3	1/2	1

- **Solution:** Random Surfer model
- Surfer starts a random page P_1 and moves to subsequent pages $P_2, P_3, \cdots \cdots \cdots, P_m$ in random order following the hyperlink network
- Upon landing at a page P_i, the surfer choose either of the following
  - a) With certain probability β, it continues in its course of following hyperlinks
  - b) With probability $(1-\beta)$, jump to any random page P_j in the *teleportation network*

Spider Trap

- Teleportation Network



Adjacency Matrix

	A	B	C
A	1	1	1
B	1	1	1
C	1	1	1

Transition Matrix

	A	B	C
A	1/3	1/3	1/3
B	1/3	1/3	1/3
C	1/3	1/3	1/3

Spider Trap

Lets denote uniform transition matrix by U and hyperlink transition matrix by M

Transition Matrix

M =

	A	B	C
A	1/3	1/2	0
B	1/3	0	0
C	1/3	1/2	1

Uniform Transition Matrix

U =

	A	B	C
A	1/3	1/3	1/3
B	1/3	1/3	1/3
C	1/3	1/3	1/3

Our old page rank formula :

r = Mr

Where r→ rank vector

Now lets include U in the formula

r = Mr + Ur

Updated Page Rank Algorithm

- Note that Mr is a column vector which sums up to 1 and Ur is also a column vector which sums up to 1
- Therefore, the rank vector r will have a sum of 2, which is inconvenient
- We will change it in such a way that it sums up to 1.
- So lets write: $r = 0.5\ Mr + 0.5\ Ur$
- This model assigns equal importance to teleportation matrix and hyperlink matrix
- We can also change the importance assigned to each matrix in following way: $r = 0.7\ Mr + 0.3\ Ur$
- In above equation we have assigned 70% importance to hyperlink matrix and 30 % importance to teleportation matrix.

Updated Page Rank Algorithm

- In the generalized way we can write:

$$r = \beta Mr + (1-\beta) Ur$$

- Where $\beta \rightarrow$ probability that a random surfer will follow the hyperlink network

$(1-\beta) \rightarrow$ probability that a random surfer will teleport to a random web page

- This β is called the damping factor.
- Its value is around 0.85 (empirically determined)

Updated Page Rank Algorithm

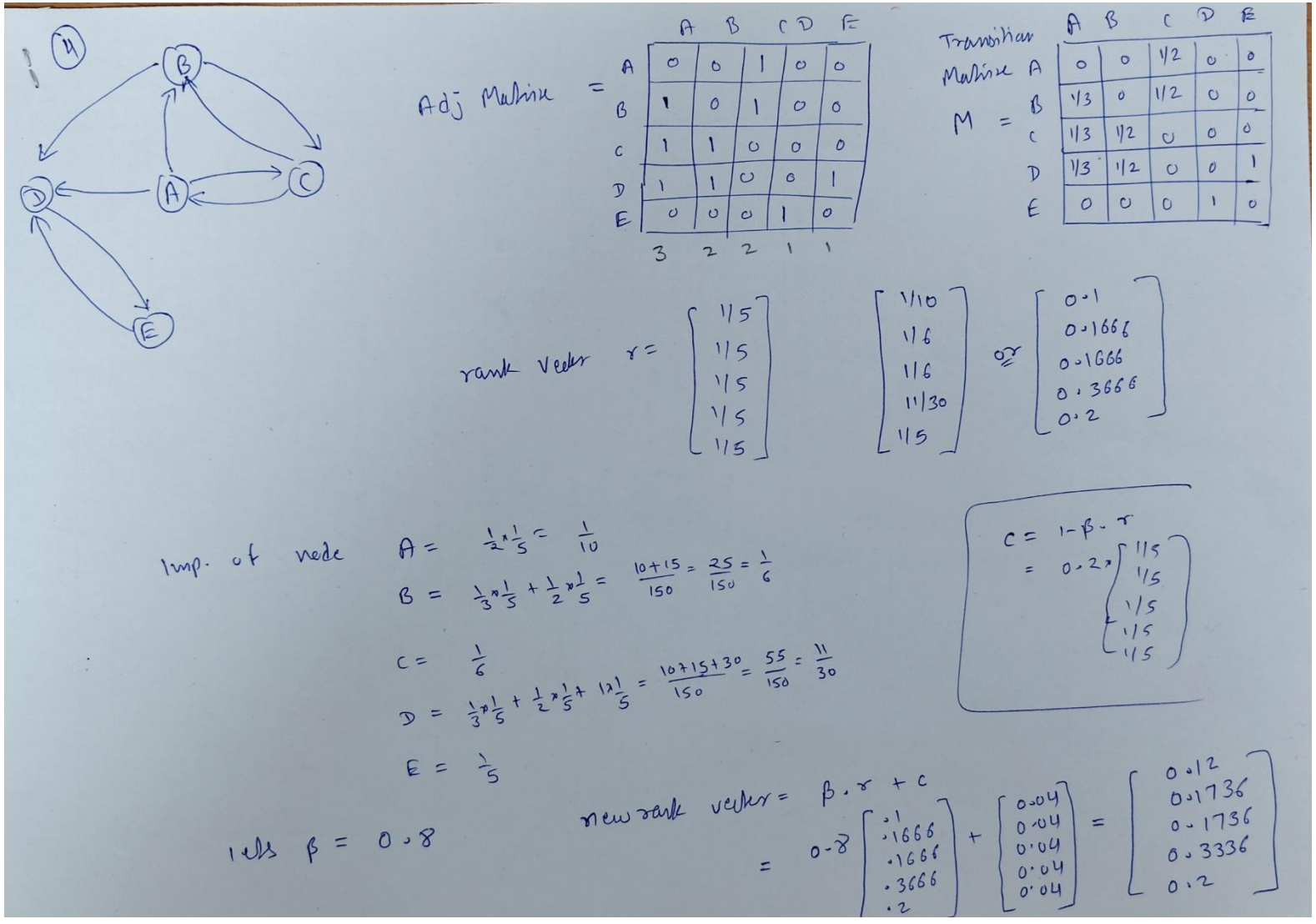
- Note that in above equation we have 2 matrix multiplication to be performed in each iteration
- The matrices will be very huge in size, so lets try to reduce the computational cost
- Notice that U is a uniform matrix, so its U.r will always be a uniform rank vector
- Therefore, we can replace $(1-\beta).Ur$ with a constant $\hat{c}$

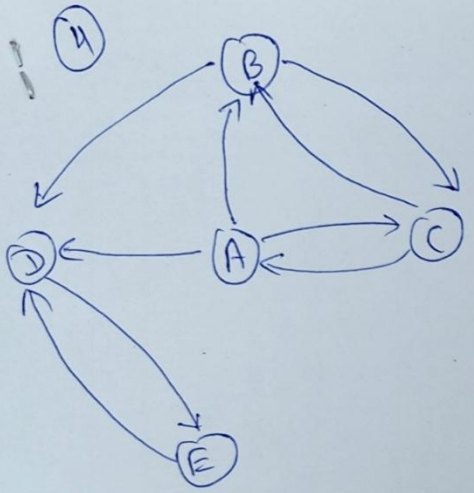
$$\mathbf{r} = \boldsymbol{\beta}.\mathbf{Mr} + \hat{\boldsymbol{c}}$$

where, $\hat{c} = (1-\beta).Ur$

Updated Page Rank Algorithm

- Example:





Adj Matrix =

	A	B	C	D	E
A	0	0	1	0	0
B	1	0	1	0	0
C	1	1	0	0	0
D	1	1	0	0	1
E	0	0	0	1	0
	3	2	2	1	1

Transition Matrix

$M =$

	A	B	C	D	E
A	0	0	1/2	0	0
B	1/3	0	1/2	0	0
C	1/3	1/2	0	0	0
D	1/3	1/2	0	0	1
E	0	0	0	1	0

rank vector $r = \begin{bmatrix} 1/5 \\ 1/5 \\ 1/5 \\ 1/5 \\ 1/5 \end{bmatrix} \quad \begin{bmatrix} 1/10 \\ 1/6 \\ 1/6 \\ 11/30 \\ 1/5 \end{bmatrix}$ or $\begin{bmatrix} 0.1 \\ 0.1666 \\ 0.1666 \\ 0.3666 \\ 0.2 \end{bmatrix}$

Imp. of node

$A = \frac{1}{2} \times \frac{1}{5} = \frac{1}{10}$

$B = \frac{1}{3} \times \frac{1}{5} + \frac{1}{2} \times \frac{1}{5} = \frac{10+15}{150} = \frac{25}{150} = \frac{1}{6}$

$C = \frac{1}{6}$

$D = \frac{1}{3} \times \frac{1}{5} + \frac{1}{2} \times \frac{1}{5} + 1 \times \frac{1}{5} = \frac{10+15+30}{150} = \frac{55}{150} = \frac{11}{30}$

$E = \frac{1}{5}$

$$\boxed{c = 1 - \beta \cdot r = 0.2 \times \begin{bmatrix} 1/5 \\ 1/5 \\ 1/5 \\ 1/5 \\ 1/5 \end{bmatrix}}$$

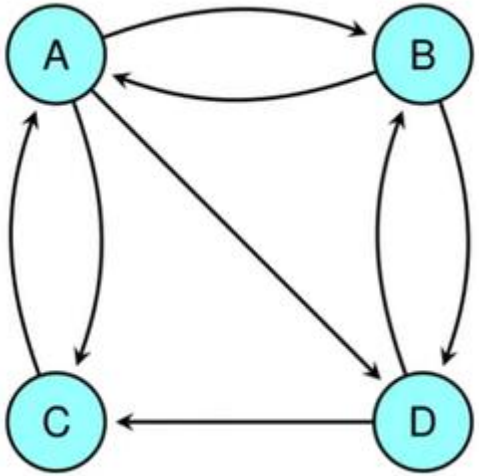
let $\beta = 0.8$

new rank vector $= \beta \cdot r + c$

$$= 0.8 \begin{bmatrix} .1 \\ .1666 \\ .1666 \\ .3666 \\ .2 \end{bmatrix} + \begin{bmatrix} 0.04 \\ 0.04 \\ 0.04 \\ 0.04 \\ 0.04 \end{bmatrix} = \begin{bmatrix} 0.12 \\ 0.1736 \\ 0.1736 \\ 0.3336 \\ 0.2 \end{bmatrix}$$

Katz Centrality

Katz centrality

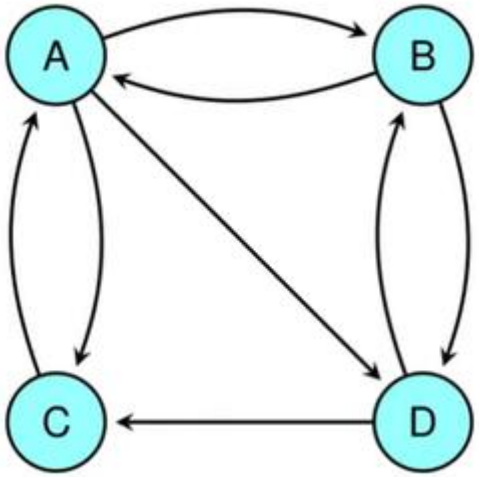


Adjacency matrix:

$$\mathbf{M} = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

(number of paths of length 1)

Katz centrality



Adjacency matrix:

$$\mathbf{M} = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

(number of paths of length 1)

Matrix multiplication can be used to compute the number of paths between two nodes:

$$\mathbf{MM} = \begin{bmatrix} 2 & 0 & 0 & 2 \\ 1 & 2 & 1 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$ (number of length-2 paths)

$$\mathbf{MMM} = \begin{bmatrix} 2 & 4 & 2 & 0 \\ 3 & 1 & 1 & 3 \\ 3 & 1 & 1 & 3 \\ 3 & 2 & 1 & 2 \end{bmatrix}$$ (number of length-3 paths)

Katz Centrality

- Can we use path counting to compute centrality?
- Idea: A node is central if many paths lead to it.
- Problem: there can be an infinite number of (directed) paths in a graph with cycles.
- Solution: (Leo Katz, 1953) Penalize longer paths using (rapidly shrinking) powers of some attenuation factor $0<\alpha<1$

Katz Centrality

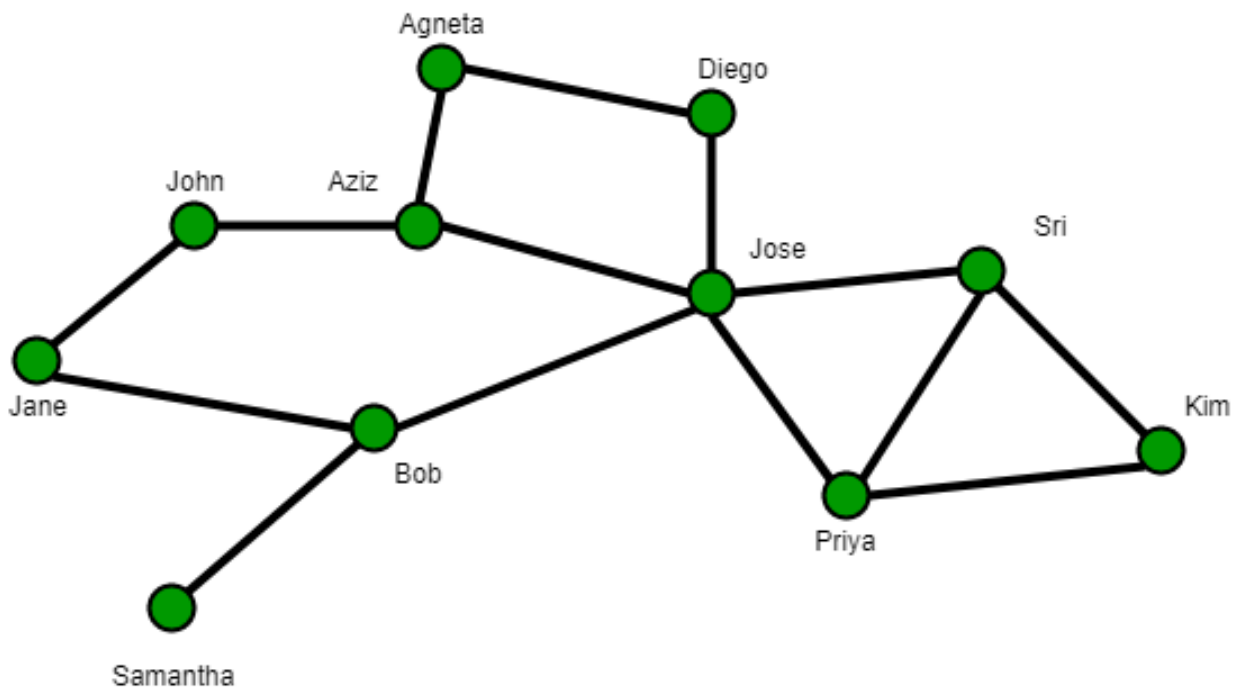
Definition 13.8: Given an $n \times n$ column-based adjacency matrix $\mathbf{M}$ and a real number $0 < \alpha < 1$, the (in-bound/receiving) Katz centrality of a node $i \leq n$ is the result of the limit

$$\sum_{k \geq 1} \sum_{j=1}^{n} \alpha^k (\mathbf{M}^k)_{ij},$$

i.e., sum of the α^k-weighted sums of the i-th row in matrix $\mathbf{M}^k$.

Note: For this to work, the α must not be too large. Precisely, we should have $\alpha < 1/|r|$, where r is the eigenvalue of $\mathbf{M}$ with the largest absolute value.

Katz Centrality



- Let us consider the influence of ***Jose*** in the network, and also let the attenuation factor be α, $0 < \alpha < 1$
- Immediate neighbours of Jose are ***Diego, Aziz, Bob, Priya,*** and ***Sri.*** Influence of these neighbours on Jose would be attenuated at a factor of α
- Second order neighbours of Jose are ***Agneta, John, Samantha,*** and ***Kim.*** Influence of these neighbours on Jose would be attenuated at a factor of α^2
- The (only) third order neighbour of Jose is ***Jane.*** Influence of these neighbours on Jose would be attenuated at a factor of α^3

Katz Centrality

Katz overcomes some problems

- Dead ends do not drain all weight from the remaining graph (Katz centrality is meaningful on acyclic graphs)
- Spider traps do not drain all weight from the remaining graph (but they retain some centrality advantage)

The method is sensitive to the choice of the attenuation factor:

- Large values lead us to consider long paths, making results somewhat similar to eigenvector centrality.
- Small values focus on (very) short paths, making results more similar to in-degree centrality.